

Tax Calculator Problem

GlobalNext Solutions, a rapidly growing IT company, employs a diverse workforce ranging from entry-level developers to senior executives. The HR department wants to streamline the tax calculation process for employees under the New Tax Regime (2023). They've decided to build a tax calculation program that computes salaries, taxes, and net incomes while ensuring compliance with the latest tax laws.

As a software developer in GlobalNext's HR-Tech team, you are tasked with developing this program. The system should process employee salary details, validate inputs, calculate taxes, and generate detailed reports.

The program should:

1. Accept employee details, including monthly salary components.
2. Calculate gross and taxable income according to the New Tax Regime (2023).
3. Compute the tax payable using the appropriate tax slabs.
4. Apply any applicable standard deductions and rebates.
5. Generate reports detailing gross salary, taxable income, tax payable, and net salary.

Level 1: Basic Input and Salary Calculation

Objective: Capture employee details and calculate the gross salary. Tasks:

- Accept the following inputs for an employee:
 - o Name
 - o EmpID
 - o Basic Monthly Salary
 - o Special Allowances (Monthly)
 - o Bonus Percentage (Annual Bonus as % of Gross Salary)
- Calculate:
 - o $\text{Gross Monthly Salary} = \text{Basic Salary} + \text{Special Allowances}$
 - o $\text{Annual Gross Salary} = (\text{Gross Monthly Salary} \times 12) + \text{Bonus}$
- Output:
 - o Display the employee details, gross monthly salary, and annual gross salary

Level 2: Taxable Income Calculation

Objective: Calculate taxable income after standard deductions. Tasks:

- Deduct a Standard Deduction of ₹50,000 from the annual gross salary.
- Compute the Taxable Income and display all intermediate calculations.

Output: Display gross salary, standard deduction and taxable income.

Level 3: Tax and Rebate Calculation

Objective: Compute tax payable using the New Tax Regime (2023) slabs. Tasks:

1. Calculate tax based on the following slabs: o ₹0- ₹3,00,000: 0% o ₹3,00,001- ₹6,00,000: 5% o ₹6,00,001- ₹9,00,000: 10% o ₹9,00,001- ₹12,00,000: 15% o ₹12,00,001- ₹15,00,000: 20% o Above ₹15,00,000: 30%

2. Apply Section 87A Rebate:

o Taxable income $\leq$ ₹7,00,000 $\rightarrow$ 100% rebate (tax payable = ₹0).

3. Add a 4% Health and Education Cess to the calculated tax. Output:

• Display a detailed tax breakdown, including slabs, cess, and total tax payable.

Level 4: Net Salary Calculation

Objective: Calculate annual net salary after tax deductions. Tasks:

1. Compute Net Salary = Annual Gross Salary- Total Tax Payable.

2. Display:

o Annual Gross Salary

o Total Tax Payable (including cess)

o Annual Net Salary

Level 5: Report Generation

Objective: Generate a detailed report for employees. Tasks:

1. Summarize all computed details:

o Employee Details (Name, EmpID)

o Gross Monthly Salary

o Annual Gross Salary

o Taxable Income

o Tax Payable (with breakdown)

o Annual Net Salary

2. Format the output as a report for better readability.

Output:

• Provide a clean, tabular report for employees.

Level 6: Input Validation Rules

Objective: Validate all inputs to ensure accuracy and correctness. Validation Rules:

1. Employee Details:

- o Name: Non-empty, alphabets only, max 50 characters.
- o EmpID: Alphanumeric, 5–10 characters.

2. Salary Inputs:

- o Basic Salary: Positive number, max ₹1,00,00,000.
- o Special Allowances: Non-negative, max ₹1,00,00,000.
- o Bonus Percentage: Numeric value, 0–100.

3. Derived Calculations:

- o Gross Monthly Salary must be greater than zero.
- o Annual Gross Salary should not exceed realistic values.

4. General:

- o Reject invalid inputs with a clear error message.
- o Provide re-entry prompts for invalid data.

Output:

- Indicate if any inputs are invalid and prompt for correction.